

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 22

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable • If it cannot be adjusted the percentage error should be recorded so the results can be corrected	2
01.2	<i>Any 3 from:</i> • Droppings: diet, gender, territory size • Owl pellets: diet • Feeding marks (eg chewed hazel nuts): presence of species such as dormice • Nests / burrows: presence and breeding • Tracks / footprints: presence, movements • Territorial marks (scratching posts): presence and territory	3
01.3	• Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species • The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen	2
01.4	<i>Any 3 from:</i> • Marked individuals mix freely with the rest of the population • Marks do not fall off or wear away, and marking does not affect survival • The population does not change during the study (no births, deaths or migration) • All individuals have an equal chance of being caught / animals are mobile and can be caught	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
02.1	• Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids • Desired characteristics introduced without linked unwanted ones; faster	2

Question	Marking guidance	Marks
02.2	• Bt: reduces insecticide spraying so fewer non-target deaths, lower cost / energy • Roundup Ready: weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage • Golden rice: provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa	3
02.3	• Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) • Antibiotic-resistance marker genes could transfer to pathogenic bacteria; possible allergies; gene transfer to gut bacteria • Detection in the human food chain from crops approved only for animal feed • Economic: seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs; herbicide-resistant crops may increase herbicide use	4
02.4	• Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	
03.1	• Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis; human processes usually use high energy-density non-renewable fossil fuels • Natural processes occur at low temperatures because enzymes lower the activation energy of reactions; human processes need high temperatures and pressures • Natural processes produce no toxic wastes / their products are raw materials for other processes; industrial processes emit CO₂ and NO_x • So natural fixation is sustainable indefinitely with a low carbon footprint	4
03.2	<i>Any 3 from:</i> • Hydrological cycle (evaporation, precipitation) • Carbon cycle / photosynthesis • Atmospheric circulation / winds • Thermohaline circulation (density differences from heating / evaporation) • Decomposition	3
03.3	• Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting • Anaerobic digestion and composting treat wastes; enzymes in industry (biological washing powders, biofuels from algae / microorganisms) • Heat pumps and solar thermal capture low-density heat; low-temperature manufacturing using biological polymers; fuel cells instead of combustion	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
04.1	• Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala); increases evaporation • Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase • Evapotranspiration changes: increased in irrigated arid areas, decreased where forests were cleared	3

Question	Marking guidance	Marks
04.2	- Seasonal / local: avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes - Organic: uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species; free-range and Fairtrade also accepted for welfare / social reasons	4
04.3	- Food shortages after 1945 (dependence on US food aid); farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock - Guaranteed market / price support: government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output - Consequences: loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution; later food surpluses (butter mountains) that could not be sold without harming LEDC producers	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
05.1	- B: frost-free period (150 days) exceeds the 120 needed whereas A (110) is too short - late frosts would kill seedlings - B is warmer / longer thermal growing season because south-facing slopes receive more insolation; valley floors are frost pockets where cold dense air collects - But B has lower rainfall and a steeper slope (erosion / machinery difficulties) so irrigation / contour cultivation may be needed	3
05.2	<i>Any 3 from:</i> - Greenhouses / polytunnels heated by solar gain or burning gas / paraffin (ventilated to prevent overheating) - Woven plastic cloches / fleece over rows to retain warm air near the ground - Burning oil / gas or using large fans in orchards to prevent frost damage to blossom - Site selection (aspect, avoiding frost pockets); heated buildings and shelter for livestock	3
05.3	- Mammals and birds use food energy to maintain a constant body temperature (thermoregulation) - Reducing heat loss (shelter, heated buildings) means less food energy is used for heat and more is available for growth	2
05.4	- Long days increase milk production and egg laying; poultry grow best with short days; sheep mate as days shorten in autumn - Artificial lighting extends the growing season for greenhouse crops or simulates autumn in spring to give sheep a second mating / lambing season	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-309; AQA spec 3.5.1.1.	
06.1	B	1

Question	Marking guidance	Marks
06.2	<i>Any 4 from:</i> • Harvesting no faster than the MSY - resource never depleted • Mixed species plantations - more food choices, relationships and niches • Indigenous species - support indigenous insects, birds • Mixed age structure - varied light, temperature, wind so more species; continuous cover and yield • Selective logging - small clearings, less disturbance, long-term timber benefit • Retention of dead wood, buffer zones, corridors	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
07.1	• General physiological solvent: most biological reactions take place between substances dissolved in water • Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes • High specific heat capacity moderates temperature changes; evaporation absorbs heat (cooling) • Anomalous expansion: ice floats so water bodies do not freeze solid; absorbs UV protecting early marine life; provides aquatic habitats	4
07.2	• From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen; at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) • Oxygen in the stratosphere absorbed UV, forming ozone (O₃) in a dynamic equilibrium of O, O₂ and O₃ • The ozone layer reduced UV reaching the surface so organisms could survive out of the water • Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved; many anaerobic archaea died out	4
07.3	• Photosynthesis removed CO₂ (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels • This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
08.1	• Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands; HEP schemes alter river levels flooding sandbank nests of turtles / birds; species with narrow tolerance (specialists) affected first; sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade • Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae • Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish)	6
08.2	• Reduces light penetration so submerged plants cannot photosynthesise • Blocks the gills of filter feeders such as bivalve molluscs; sediment smothers the bed	2

Question	Marking guidance	Marks
08.3	• Rare species often have very specialised habitat requirements / narrow ranges of tolerance • A small change moves conditions outside their range while generalists survive; small populations cannot recover	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	
09.1	• Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud • Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores); salt excretion • Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration	3
09.2	• Coastal erosion / storm protection: roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact) • Fisheries: nursery grounds where dense roots protect young fish that later live on reefs / open sea • Protection of coral reefs: slow river water so suspended solids settle and nutrients are absorbed before reaching reefs • Resources: timber, fuel; medicines (antimicrobials against MRSA, cyanobacterium for lung cancer); high biodiversity (crabs, shrimps, sponges, birds)	4
09.3	• Clearance for aquaculture (shrimp ponds), urban development, ports, tourism; pollution; sea level rise / climate change; destruction of coral reefs that shelter the coast • Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015) • Replanting and natural regeneration; control of damaging activities; Ramsar sites	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • In-situ: species remain in natural conditions with the whole community and inter-species relationships; protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • Limits of in-situ: external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • Ex-situ: captive breeding when in-situ is impossible or insufficient; selection criteria; difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation); techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning); seed banks; successes (Hawaiian goose, Iberian lynx, red kite, large blue) • Limits of ex-situ: expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • Judgement: ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to; both needed, in-situ fundamental	9
10.2	• Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Ecological features: cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web; specific abiotic requirements (temperature, light, salinity, low turbidity) • Importance: quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • Vulnerability: narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching; slow growth so slow recovery; physical damage; overfishing and food-web effects (crown of thorns); introduced species; dependence on mangroves and seagrass • Conservation: MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25